

Reproducibility

Reproducibility in computational research has well-documented prerequisites: open data, open code, and a deterministic build that can be re-run from scratch (Peng 2011). The bundled `manuscript/config.yaml` is intentionally configured to satisfy all three for **strict reproducibility**:

# Reproducibility I

1. `project_config.search.sources`: `[local]` consumes `data/corpus.json`, which is a curated and committed JSON corpus. No network is required to run the pipeline.
2. `search.cache_dir`: `output/search/cache` writes deterministic JSON cache files; running the same query twice produces a byte-identical artifact tree (modulo timestamp metadata in the cache file itself).
3. `enrichment.fetch_abstracts`: `true` reads abstracts directly from the corpus when present; no network fetch is required.
4. `enrichment.fetch_fulltext`: `false` is the default — full-text fetching is opt-in and gated behind the optional `pypdf` dependency (`uv sync --group rendering`).
5. `llm.enabled`: `false` is the default — the LLM stage is opt-in and requires a running `ollama serve`. When enabled, `seed`: `42` and `temperature`: `0.0` are pinned.

## Switching to live search I

Replace the sources list with the desired backend set:

```
search:  
  query: "your topic"  
  sources: [arxiv, crossref]  
  crossref_mailto: "you@example.org"
```

## Switching to live search I

A first live run populates `output/search/cache/`. Commit that directory to the repo and the pipeline becomes reproducible across machines without further configuration changes.

## Determinism guarantees I

The full determinism contract is itemised in tbl. 1: every pipeline stage is annotated as fully, conditionally, or non-deterministic, with an explicit mechanism column so reviewers can audit each row independently.

# Determinism guarantees I

Table 1: Determinism contract by pipeline stage. Cached stages are byte-stable across reruns; live stages depend on the upstream source and are pinned to the cache file once a successful run completes.

| Stage                   | Deterministic?                  | Mechanism                                                                                                          |
|-------------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Search (cached hit)     | yes                             | SearchCache JSON files                                                                                             |
| Search (cache miss)     | no                              | live API                                                                                                           |
| Dedup / merge           | yes                             | DOI / arXiv-id canonical keys;<br>tie-break by score then year                                                     |
| Citation-key generation | yes                             | unicode folding + stop-word skip;<br>collision suffix is deterministic                                             |
| BibTeX writer           | yes                             | byte-stable format pinning (verified<br>by tests/infra_tests/reference/)                                           |
| Abstract fetch          | yes (cached) / no (live)        | per-paper <safe_id>.txt cache                                                                                      |
| Fulltext fetch          | yes (cached) / mostly (live)    | per-paper <safe_id>.{pdf,txt}<br>cache; live fetch's pypdf text<br>extraction is not bit-stable across<br>versions |
| LLM synthesis           | mostly                          | seed=42, temperature=0.0; Ollama<br>deterministic up to its own minor<br>variance                                  |
| Figure generation       | yes (within Matplotlib version) | fixed palette, fixed bin width, no<br>random subsampling                                                           |

## Verifying reproducibility locally I

```
# Run twice; nothing in output/ should diff except the cache  
uv run python projects/templates/template_search_project/search.py  
mv projects/templates/template_search_project/output projects/templates/template_search_project/output_first  
uv run python projects/templates/template_search_project/search.py  
diff -ru \  
    projects/templates/template_search_project/output_first  
    projects/templates/template_search_project/output/corpus
```



## Verifying reproducibility locally I

The only expected differences are inside `output/search/cache/search_*.json`, where `_cached_at` is wall-clock time at write.

## Limitations I

The reproducibility contract enumerated in `sec. sec:reproducibility` (items 1–5 and `tbl. 1`) does not eliminate the following well-defined sources of non-reproducibility, which are surfaced here so reviewers can audit them explicitly rather than inferring from the contract table:

## Limitations I

- ▶ **Live search drift.** When `project_config.search.sources` includes `arxiv` or `crossref`, the first cache-miss invocation hits the live API; the cached JSON freezes that response, but two cold-start clones running on different days will see different paper sets. Pin a `LocalBackend` corpus or commit `output/search/cache/` to break this dependency.
- ▶ **pypdf version drift.** The fulltext fetcher uses `pypdf` to extract text from a downloaded PDF. `pypdf`'s text-extraction algorithm is not bit-stable across major versions; upgrading `pypdf` can produce different `<safe_id>.txt` cache contents from the same source PDF. The PDF bytes themselves are bit-stable so the cache freezes the inputs, not the extraction.
- ▶ **Ollama version drift.** Pinning `seed=42` and `temperature=0.0` controls Ollama's sampling, but the model weights, tokenizer, and template can change between Ollama releases. Document the Ollama version alongside `config.llm.model` when archiving a run for replication.
- ▶ **Paperclip backend status.** The `paperclip` backend is opt-in and the production endpoint may reject the adapter's request method; the run records the response in `SearchResult.errors[paperclip]` and continues. Treat `paperclip` results as advisory until the upstream service stabilises.

## Limitations II

- ▶ **External backend behaviour outside this project's control.** arXiv and Crossref are the source of truth; this project is faithful to whatever they return. A retraction, metadata fix, or DOI assignment upstream will alter the cache on the next cold-start invocation.

## Limitations I

These limitations bound *what* the cache + seed + corpus pinning achieves. Inside those bounds, the contract in tbl. 1 is total: every cached pipeline stage is byte-stable across reruns (verified by `tests/test_pipeline.py::TestRunLiteraturePipeline::test_byte_identities_across_reruns`).